

# Theory of Computation

Submission link: <https://forms.gle/9QfcWiqewGmCcvG36>

## Exercise 15: (Turing Machine)

6301134 Chayut Liewlam

\*1. Find the language of Turing Machine M.

$$M = (\{q_0, q_1, q_2, q_3, q_4, q_f\}, \{a, b\}, \{a, b, x, y, \square\}, \delta, q_0, \square, \{q_f\})$$

$$\delta: \delta(q_0, x) = (q_0, x, R),$$

$$\delta(q_0, y) = (q_0, y, R),$$

$$\delta(q_0, a) = (q_1, x, R),$$

$$\delta(q_0, b) = (q_3, y, R),$$

$$\delta(q_0, \square) = (q_f, \square, L),$$

$$\delta(q_1, a) = (q_1, a, R),$$

$$\delta(q_1, y) = (q_1, y, R),$$

$$\delta(q_1, b) = (q_2, y, L),$$

$$\delta(q_2, a) = (q_2, a, L),$$

$$\delta(q_2, y) = (q_2, y, L),$$

$$\delta(q_2, x) = (q_0, x, R),$$

$$\delta(q_3, a) = (q_4, x, L),$$

$$\delta(q_3, b) = (q_3, b, R),$$

$$\delta(q_3, x) = (q_3, x, R),$$

$$\delta(q_4, b) = (q_4, b, L),$$

$$\delta(q_4, x) = (q_4, x, L),$$

$$\delta(q_4, y) = (q_0, y, R)$$



